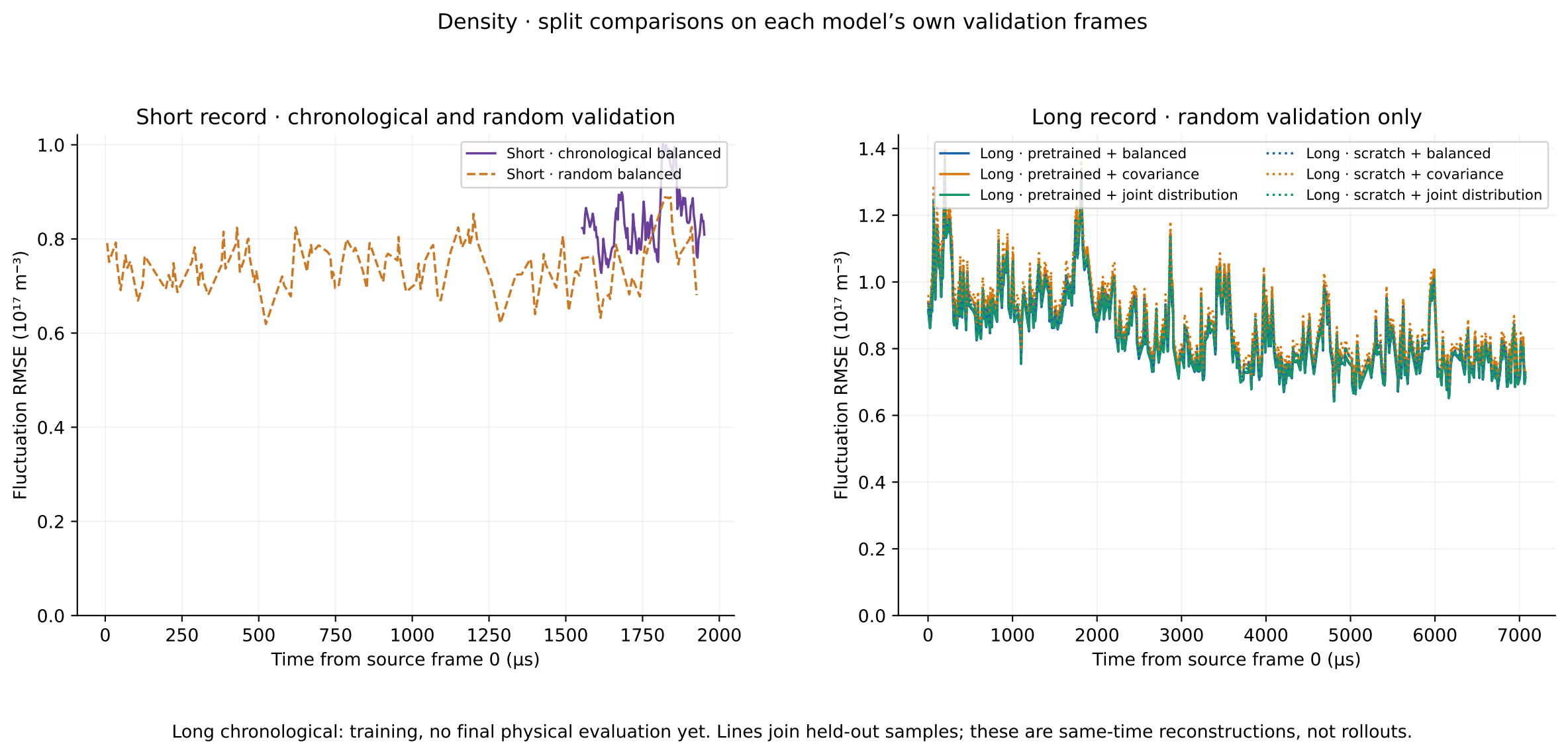
Density · split comparisons on each model’s own validation frames
Short record · chronological and random validation
Short · chronological balanced
Short · random balanced
Fluctuation RMSE ($10^{17}$ m$^{-3}$)
Time from source frame 0 (μs)
Long record · random validation only
Long · pretrained + balanced
Long · pretrained + covariance
Long · pretrained + joint distribution
Long · scratch + balanced
Long · scratch + covariance
Long · scratch + joint distribution
Fluctuation RMSE ($10^{17}$ m$^{-3}$)
Time from source frame 0 (μs)
Long chronological: training, no final physical evaluation yet. Lines join held-out samples; these are same-time reconstructions, not rollouts.